SALES DATA ANALYSIS

Project Summary Report

Python + Pandas | Submission Summary

A concise report of the completed data cleaning, analysis, reporting, charting, and testing work.

Project Objective

The project processes raw sales records from sales.csv. It identifies data-quality issues, creates a clean dataset, calculates sales values, generates reports, saves CSV outputs, creates charts, and includes basic automated checks.

Key Results

Metric	Verified result
Raw sales records	123
Final clean records	114
Rows removed during cleaning	9
Total sales	229,905.00
Average order value	2,016.71
Top product	Desk Lamp - 76,500.00
Automated tests	2 passed

Project Flow

sales.csv -> Load with Pandas -> Inspect and validate -> Clean -> Transform -> Analyse -> Export reports and charts

Data Processing and Technologies

Input CSV

sales.csv contains 123 sales records and 8 columns: order_id, order_date, customer, product, category, quantity, price, and city. Each row represents one sales record.

Data-quality issue	Finding	Handling
Exact duplicate rows	3	Removed with drop_duplicates().
Missing customer / city	1 / 1	Replaced with Unknown.
Missing price	1	Removed.
Missing or invalid date	2	Removed after date conversion.
Zero or negative quantity	2	Removed using quantity > 0.
Zero or negative price	1	Removed using price > 0.

Cleaning and Transformation

- A cleaned copy is created after duplicate removal.
- Customer and city gaps are filled with Unknown.
- Dates are converted with pd.to_datetime(..., errors="coerce", format="mixed").
- Unusable date, missing price, and non-positive quantity/price rows are removed.
- sales_amount = quantity * price, and a YYYY-MM month column is created.

Technologies and Versions

Technology	Version	Purpose
Python	3.11.4	Main language.
Pandas	2.2.3	CSV reading, cleaning, transformations, reports.
Matplotlib	3.9.2	PNG charts.
Pytest	8.4.2	Two automated checks.
CSV	File format	Input and exported summaries.

Reports, Outputs, and Code Structure

Analysis Completed

Analysis	Result
Overall performance	Total sales and average order value.
Products and customers	Top 5 products and top 5 customers by sales.
City and category	Sales totals by city and category.
Monthly trend	Monthly sales from January to July 2025.

Generated CSV Files

File	Contents
sales_summary.csv	Total sales and average order value.
customer_summary.csv	Top 5 customers by sales.
product_summary.csv	Top 5 products by sales.
city_summary.csv	Sales amount by city.
monthly_sales.csv	Sales amount by month.

Project Files

File	Responsibility
main.py	Runs the full workflow.
data_loader.py	Loads and inspects sales.csv.
data_validator.py	Shows data-quality issues.
transformations.py	Cleans data and creates sales_amount.
reports.py	Creates reports and exports CSV files.
charts.py	Creates, saves, and displays charts.
tests/test_transformations.py	Contains two basic Pytest tests.

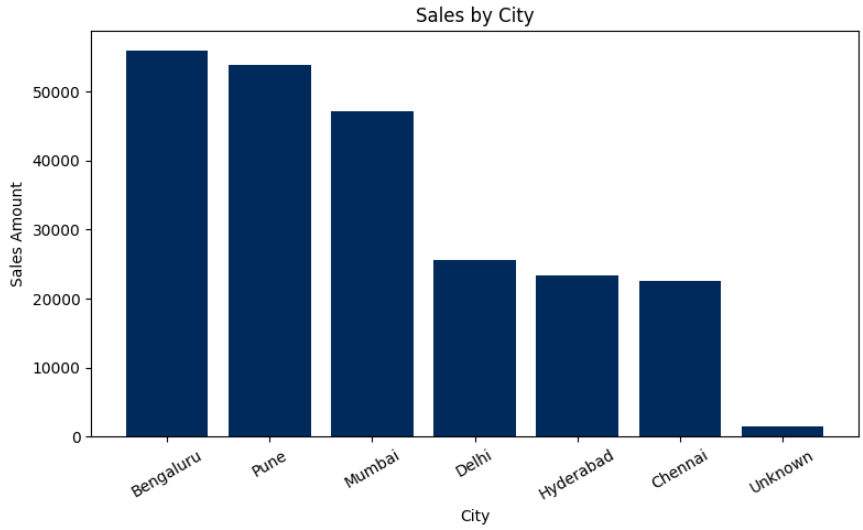
Testing and Conclusion

Two Pytest checks verify data cleaning and sales_amount calculation. The current result is 2 passed. The project uses a simple, explainable Python + Pandas workflow to turn raw sales data into reports, charts, and reusable CSV summaries.

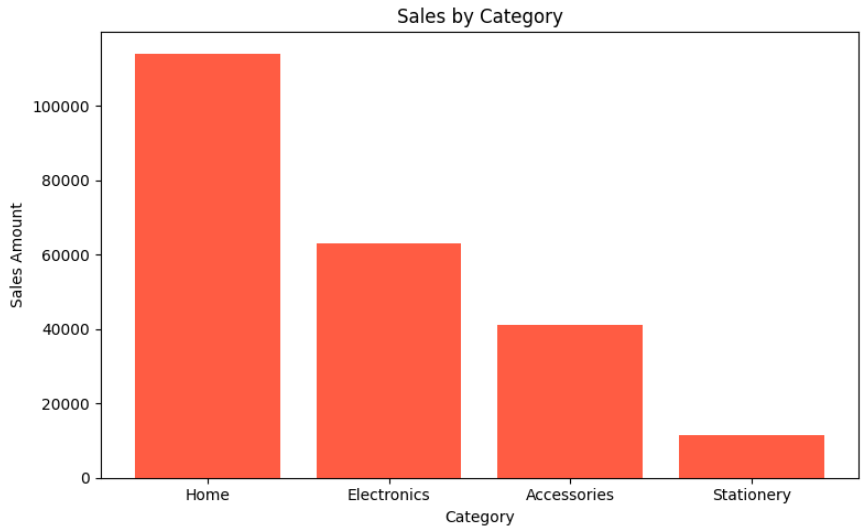
Visual Sales Summary

These charts are generated by the project and saved in output/. They use the Coforge navy-and-coral presentation palette.

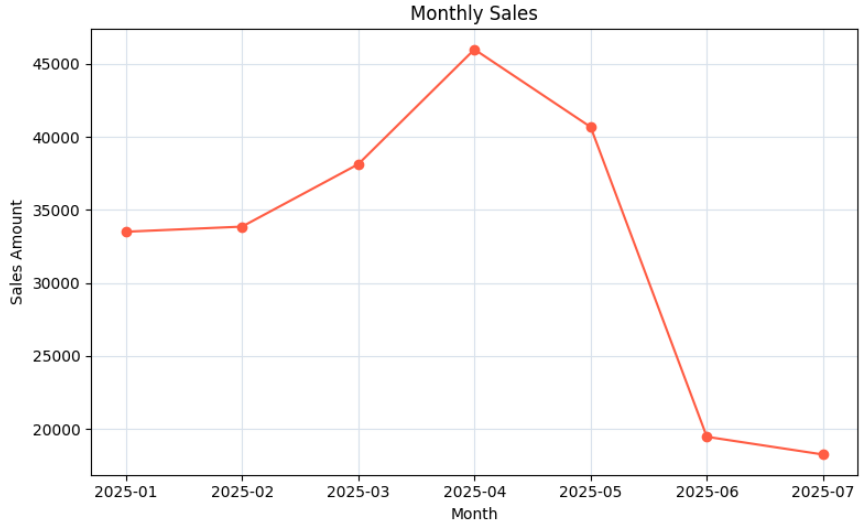
Sales by City



Sales by Category



Monthly Sales



Highlights: Bengaluru has the highest city sales (55,960.00); Home is the highest category (114,180.00); April 2025 is the highest month (46,000.00).